Assignment 1
IB3702 Mathematics for Machine Learning

Deadline: October 3, 2025

This assignment is about Calculus; this means that you have already studied all learning units from 1 to 6. Working on the problems here helps you check whether you understand the material well and are able to use calculus in various contexts.

Study advice: First of all, you should know that you are able to solve all these problems. To do that, you should study in an effective way. Most importantly, you should be **honest** with yourself; specifically, acknowledge if something is not entirely clear to you. You should **practice** with the given exercises in the workbook and from the lectures. When starting with new areas and techniques or you do not grasp them yet, you should **explore** those; see some tips in the study guide (Introduction/Section 3.1). Finally, you can (and it is highly recommended to) **talk** to others to gain more and more insights. While working on the assignment, you get a good picture about your knowledge and skills. If you are stuck with something in the assignment, you should **revisit** the study material.

Requirements: To have your assignment approved, you have to work out each problem entirely. Explain your thoughts clearly. Please do your best! Note that it is required to receive a *Pass* for both assignments in order for you to pass the whole course. Finally, use the template given on Brightspace for writing your assignment. You may include digital photos about your hand-written work in your document, provided that they are clear and readable.

Important: Although collaboration is encouraged, you have to write the assignment on your own. It is important and appreciated if you describe how you see a problem, how you get to the solution, and what kinds of challenges you faced. These aspects make your assignment richer.

Exercise 1: Sequence

Question 1.1. Given $a_0 = 1$, and the sequence be recursively defined as:

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$$

a) Compute the first five terms of the sequence.

b) Prove that the limit of this sequence is $\sqrt{2}$

Hint: The limits of a_{n+1} and a_n can both be considered as a real number L in infinity.

Question 1.2 Given $a_n = (-1)^n \cdot \frac{n}{n+1}$, determine whether the sequence a_n converges or diverges.

Exercise 2: Series

Question 2.1. Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)}$ converges.

Hint: Remember that we can interpret series also as integrals. So, if the relevant integral returns a real number, then the series is also convergent.

Question 2.2. Determine whether the following geometric series is convergent or divergent?

$$\sum_{n=0}^{\infty} (-1)^n (1/e)^n$$

(e is Euler number)

if it is convergent, to which number? if it is divergent, why?

Exercise 3: Limits

Let the function $f(x)$ be defined as:

$$f(x) = \begin{cases} 1 & \text{if } x < 0 \\ x & \text{if } 0 < x < 1 \\ 2 - x & \text{if } 1 < x < 3 \\ x - 4 & \text{if } x > 3 \end{cases}$$

1. Is it possible to define f at $x = 0$ in such a way that f becomes continuous at $x = 0$?
   If so, then we should set $f(0)$ to which value?
2. Is it possible to define f at $x = 1$ in such a way that f becomes continuous at $x = 1$?
   If so, then we should set $f(1)$ to what?
3. Is it possible to define f at $x = 3$ in such a way that f becomes continuous at $x = 3$?
   If so, then we should set $f(3)$ to what?

Exercise 4: Derivative

Question 4.1. Compute the derivative of the following functions:

1. $f(x) = \sin^2(\ln(x^2 + 1))$

2. $g(x) = 2^{\cos x} + \frac{x^2}{\sqrt{x^3}}$

Question 4.2. Let

$$f(x, y) = xsin(y) + e^{x+2y}$$

Compute the following partial derivatives:

$$\frac{\partial f}{\partial x} = ?, \quad \frac{\partial f}{\partial y} = ?, \quad \frac{\partial^2 f}{\partial x \partial y} = ?, \quad \frac{\partial^2 f}{\partial x^2} = ? \quad \frac{\partial^2 f}{\partial y \partial x} = ?$$

Exercise 5: Function Behavior

Given the function

$$f(x) = \frac{x^2}{x^2 - x - 2}.$$

Determine

- the domain,
- the zeroes (also called roots),
- the horizontal and vertical asymptotes, and
- the extreme values.

Finally, include the graph of f too.

(Hint: To determine the information requested above, you do not need to follow the same order.)

Exercise 6: Integral

Solve the following integrals:

1. $$\int_1^2 \frac{2 + 3x^2}{x} dx$$

   Hint: it can be the addition of two functions

2. $$\int x \cdot e^x dx$$

   Hint: use integration by parts